Autonomous self-burying seed carriers for aerial seeding.

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Abstract

Aerial seeding can quickly cover large and physically inaccessible areas[1] to improve soil quality and scavenge residual nitrogen in agriculture[2], and for postfire reforestation[3,4,5] and wildland restoration[6,7]. However, it suffers from low germination rates, due to the direct exposure of unburied seeds to harsh sunlight, wind and granivorous birds, as well as undesirable air humidity and temperature[1,8,9]. Here, inspired by *Erodium* seeds[10,11,12,13,14], we design and fabricate self-drilling seed carriers, turning wood veneer into highly stiff (about 4.9 GPa when dry, and about 1.3 GPa when wet) and hygromorphic bending or coiling actuators with an extremely large bending curvature (1,854 m^{-1}), 45 times larger than the values in the literature[15,16,17,18]. Our three-tailed carrier has an 80% drilling success rate on flat land after two triggering cycles, due to the beneficial resting angle (25°–30°) of its tail anchoring, whereas the natural *Erodium* seed's success rate is 0%. Our carriers can carry payloads of various sizes and contents including biofertilizers and plant seeds as large as those of whitebark pine, which are about 11 mm in length and about 72 mg. We compare data from experiments and numerical simulation to elucidate the curvature transformation and actuation mechanisms to guide the design and optimization of the seed carriers. Our system will improve the effectiveness of aerial seeding to relieve agricultural and environmental stresses, and has potential applications in energy harvesting, soft robotics and sustainable buildings.